Exp. No: 3

Force of Attraction between Two Electric Charges and Two Parallel Conductors Separated by Varying Distance

Aim:

To determine and plot the Force of attraction between two electric Charges and two parallel conductors separated by varying distance between them using MATLAB.

Objectives:

1. To compute the electrostatic force of attraction or repulsion between two point charges using Coulomb's law.

2. To compute the magnetic force per unit length between two current-carrying parallel conductors.

3. To analyze and plot the variation of force with separation distance and charge or current magnitude.

Test case (i) Determination and plotting of force between two electric charges with varying distance

```
clc;

clear;

q1 = input('Enter the charge q1=');

q2 = input('Enter the charge q2=');

epsilon = 8.854e-12;

k = 1/(4*pi*epsilon);

r = linspace(0.1,10,100);

F = k*q1*q2./(r.^2);

disp('Force at r = 1 m is:');

disp(k*q1*q2/(1^2));
```

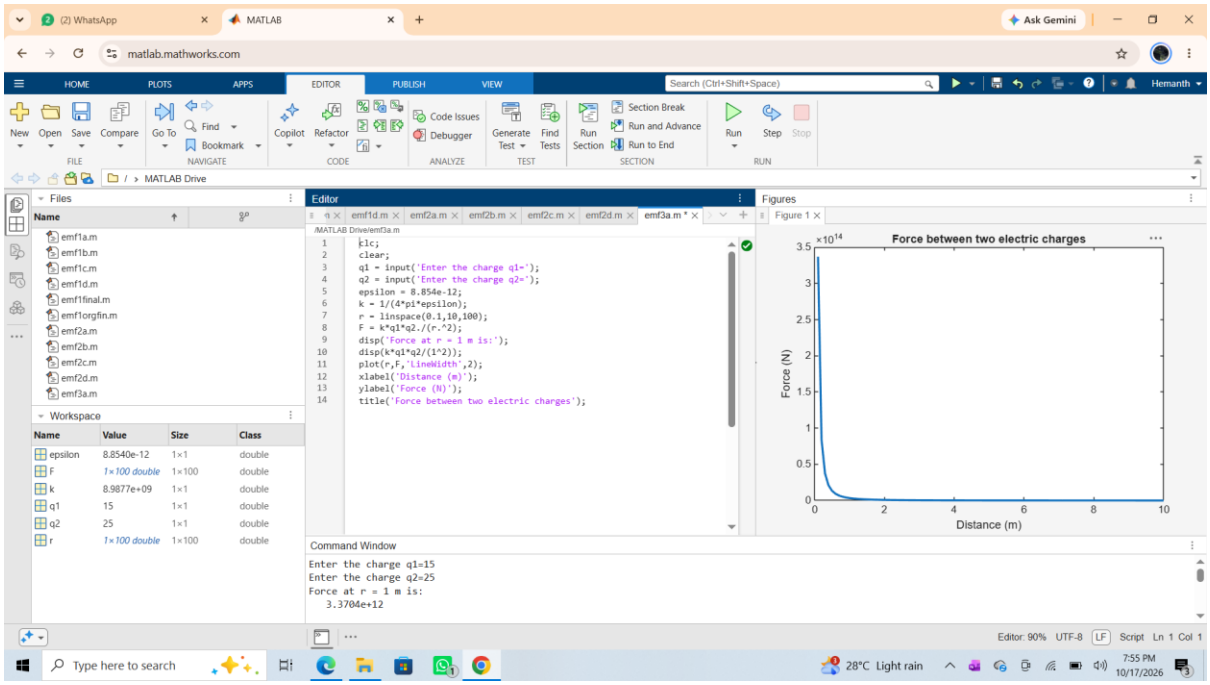
```
plot(r,F,'LineWidth',2);

xlabel('Distance (m)');

ylabel('Force (N)');

title('Force between two electric charges');
```

OUTPUT:



Test case (ii) Determination and plotting of force between two electric charges with different magnitudes Code:

```
clc;

q1 = 5e-6;

q2 = 2e-6;

epsilon = 8.854e-12;

k = 1/(4*pi*epsilon);

r = linspace(0.5,5,100);

F = k*q1*q2./(r.^2);

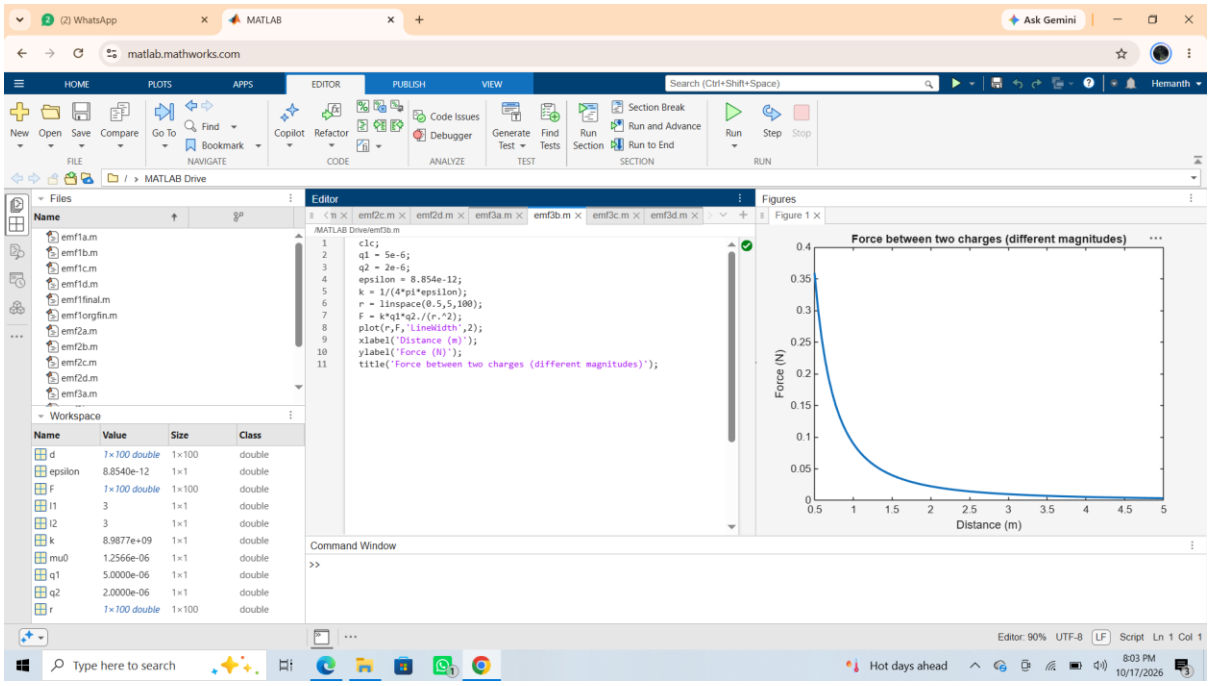
plot(r,F,'LineWidth',2);

xlabel('Distance (m)');
```

ylabel('Force (N)');

title('Force between two charges (different magnitudes)');

OUTPUT:



Test case (iii) Determination and plotting of force per unit length between two parallel conductors with varying distance

Code:

clc;

mu0 = 4*pi*1e-7;

I1 = 5;

I2 = 10;

d = linspace(0.1,10,100);

F = (mu0*I1*I2)./(2*pi*d);

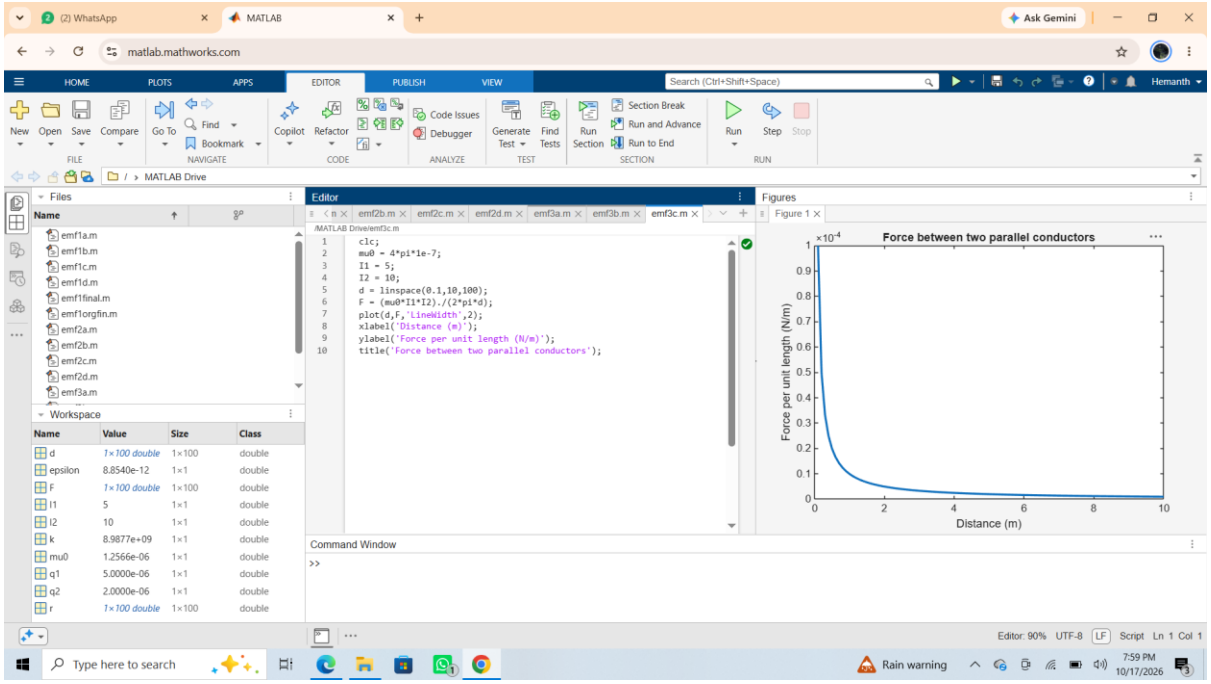
plot(d,F,'LineWidth',2);

xlabel('Distance (m)');

ylabel('Force per unit length (N/m)');

title('Force between two parallel conductors');

OUTPUT:



Test case (iv) Determination and plotting of force between two parallel conductors carrying equal currents

Code:

```
clc;

mu0 = 4*pi*1e-7;

I1 = 3;

I2 = 3;

d = linspace(0.1,2,100);

F = (mu0*I1*I2)./(2*pi*d);

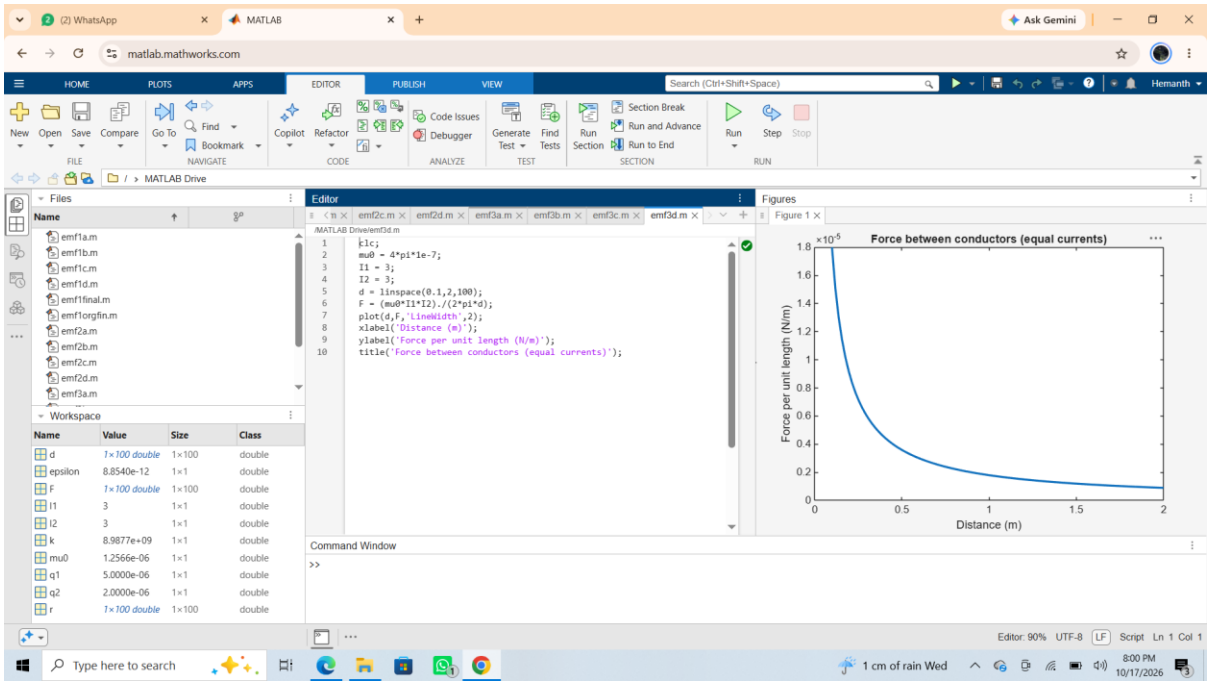
plot(d,F,'LineWidth',2);

xlabel('Distance (m)');

ylabel('Force per unit length (N/m)');

title('Force between conductors (equal currents)');
```

OUTPUT:



MATLAB Program

```
Clc;
clear;
close all;
%% Biot-Savart / Ampere-Law Magnetic Field Analysis
% Magnetic field around a long straight current-carrying conductor
%% Input Parameters I = 5; % Current in Ampere mur = 1;
% Relative permeability mu0 = 4*pi*1e-7;
% Permeability of free space (H/m) mu = mu0*mur;
% Absolute permeability % Radial distance from conductor r = linspace(0.01,1,100);

% Distance in metres %% Magnetic Field Intensity using Ampere's Law H = I ./ (2*pi*r);
%% Magnetic Flux Density B = mu .* H; m
```

```
%% Independent check: numerical Biot-Savart integration for a straight wire 100 m long (z = -50 to +50 m) H

_num = zeros(size(r)); for k = 1:length(r) am = atan(50/r(k)); alpha = linspace(-am,am,20001);

 % limiting angle of the wire ends H_num(k) = I/(4*pi*r(k))*trapz(alpha,cos(alpha));

 % H = (I/4*pi*r) * integral of cos(alpha) d(alpha) end B_num = mu .* H_num;

%% Percentage Deviation (Ampere's-law value versus Biot-Savart numerical value) H_error = abs((H_num - H) ./ H) * 100;

B_error = abs((B_num - B) ./ B) * 100;

%% Display Selected Results fprintf('\n');

 fprintf('-----------------------------------------------\n');

 fprintf(' MAGNETIC FIELD AROUND LONG STRAIGHT CONDUCTOR\n');

fprintf('-----------------------------------------------\n');

fprintf('Current I = %.2f A\n', I);

fprintf('Relative permeability mur = %.2f\n', mur);

 fprintf('-----------------------------------------------\n');

fprintf('\n r (m) H (A/m) B (T) H Error (%%) B Error (%%)\n');

fprintf('-------------------------------------------------------------------------\n');

% Display results at selected distances r_selected = [0.01 0.05 0.10 0.20 0.50 1.00];

for k = 1:length(r_selected) rr = r_selected(k); [~,kk] = min(abs(r - rr));

 HH = I/(2*pi*rr); BB = mu*HH;

fprintf('%8.2f %10.4f %10.4e %8.4f rr, HH, BB, H_error(kk), B_error(kk));

 end %% Plot H versus radial distance figure;

plot(r,H,'LineWidth',2);

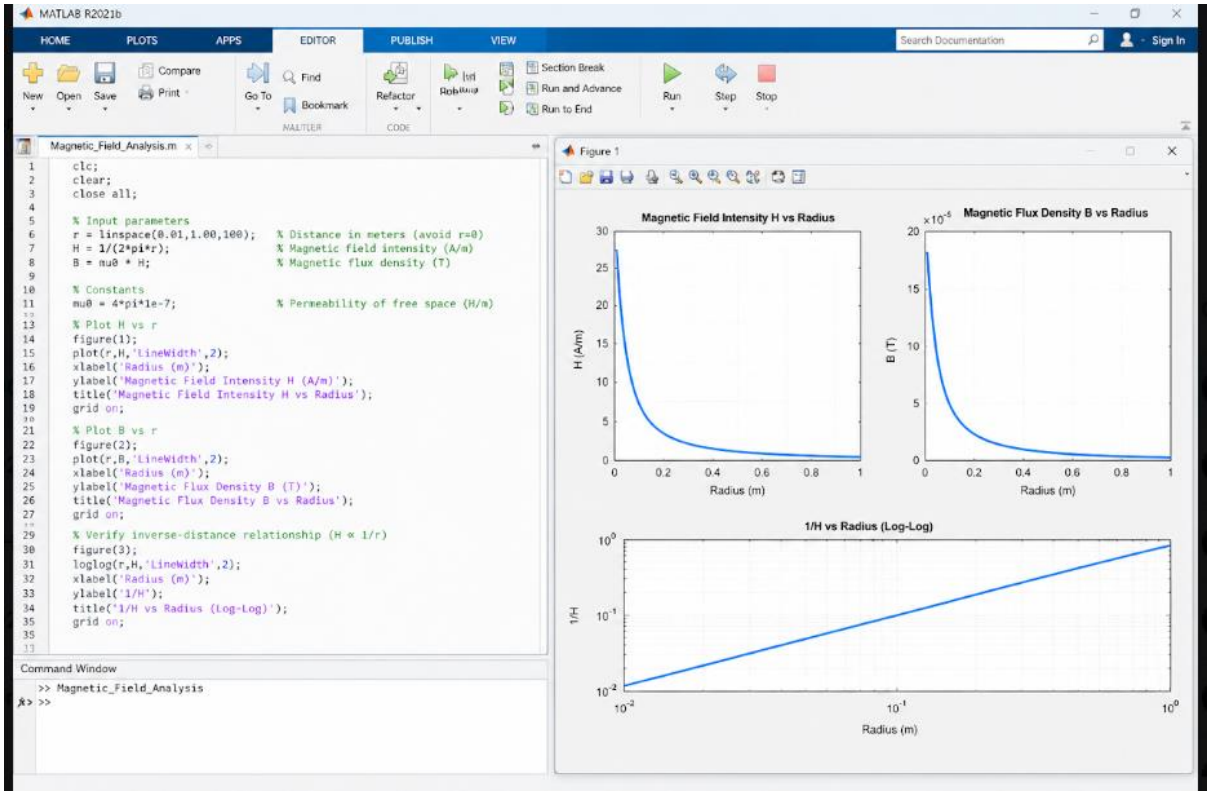
 grid on; xlabel('Radial Distance r (m)');

 ylabel('Magnetic Field Intensity H (A/m)'); %8.4f\n', ... title('Magnetic Field Intensity H versus Radial Distance');

%% Plot B versus radial distance figure;
```

```
plot(r,B,'LineWidth',2); grid on; xlabel('Radial Distance r (m)'); ylabel('Magnetic Flux Density B (T)');
 title('Magnetic Flux Density B versus Radial Distance');
 %% Verify inverse-distance relationship figure;
 loglog(r,H,'LineWidth',2); grid on; xlabel('Radial Distance r (m)'); ylabel('Magnetic Field Intensity H (A/m)');
title('Verification of H \propto 1/r');
 %% Verify B inverse-distance relationship figure;
 loglog(r,B,'LineWidth',2);
 grid on; xlabel('Radial Distance r (m)'); ylabel('Magnetic Flux Density B (T)');
 title('Verification of B \propto 1/r');
%% Calculate maximum deviation max_H_error = max(H_error); max_B_error = max(B_error);
 fprintf('\nMaximum percentage deviation in H = %.6f %%\n', max_H_error);
 fprintf('Maximum percentage deviation in B = %.6f %%\n', max_B_error);
 fprintf('\nRESULT:\n');
 fprintf('The magnetic field intensity H and magnetic flux density B\n');
fprintf('were calculated and plotted using MATLAB.\n');
 fprintf('Both H and B decrease approximately as 1/r.\n');
fprintf('The MATLAB results agree with the analytical expressions.\n');
```

OUTPUT:


MATLAB R2021b
HOME
PLOTS
APPS
EDITOR
PUBLISH
VIEW
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Sign In
New
Open
Save
Compare
Print
Go To
Find
Bookmark
Refactor
Section Break
Run and Advance
Run to End
Run
Step
Stop
Magnetic_Field_Analysis.m
clc;
clear;
close all;
% Input parameters
r = linspace(0.01,1.00,100); % Distance in meters (avoid r=0)
H = 1/(2*pi*r); % Magnetic field intensity (A/m)
B = mu0 * H; % Magnetic flux density (T)
% Constants
mu0 = 4*pi*1e-7; % Permeability of free space (H/m)
% Plot H vs r
figure(1);
plot(r,H,'LineWidth',2);
xlabel('Radius (m)');
ylabel('Magnetic Field Intensity H (A/m)');
title('Magnetic Field Intensity H vs Radius');
grid on;
% Plot B vs r
figure(2);
plot(r,B,'LineWidth',2);
xlabel('Radius (m)');
ylabel('Magnetic Flux Density B (T)');
title('Magnetic Flux Density B vs Radius');
grid on;
% Verify inverse-distance relationship (H ∝ 1/r)
figure(3);
loglog(r,H,'LineWidth',2);
xlabel('Radius (m)');
ylabel('1/H');
title('1/H vs Radius (Log-Log)');
grid on;
Command Window
>> Magnetic_Field_Analysis
>>
Figure 1
Magnetic Field Intensity H vs Radius
H (A/m)
Radius (m)
Magnetic Flux Density B vs Radius
B (T)
Radius (m)
1/H vs Radius (Log-Log)
1/H
Radius (m)

S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Identify physical quantities and force models	Identify physical quantities and force models	Correctly identifies charge/current, separation, medium and force direction for both cases.	Identifie s most quantities and models correctly.	Needs guidance in selecting the correct relation or direction.	Uses incorrect force models or quantities	25	2.5	
2	Calculate electric and magnetic forces	Calculate electric and magnetic forces	Implements Coulomb-force and parallel current force calculations correctly with units.	Calcula t ions are mostly correct with minor errors.	. Some calculations are correct but require assistance.	Force calculations are invalid or absent.	25	2.5	
3	Evaluate force versus separation	Evaluate force versus separation	Correctly varies separation, plots trends and checks theoretical distance dependence..	Trend and plots are correct with minor omissio ns.	. Limited sweep or incomplete trend interpretation.	Does not establish a valid relationship	30	3	
co14	Explain and report results	Explain and report results	. Explains attraction/repulsi on and reports assumptions, units and conclusions clearly	Conclus ion is correct with minor docume ntation gaps..	Conclusion is incomplete or partly inaccurate.	Conclusion is missing or incorrect.	20	2	
Total							100	10	